

## ***B8IT123 CA1 Group 6 Data Visualisation Project***

**Decade-long analysis on fuel mix usage by household in Ireland**

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## 1. Scope of the project

This report was based on a set of questions that defined the scope of the project and its objectives:

- Have the five most populous countries in Europe increased the usage of sustainable energy sources over the last decade?
- What about Ireland?
- What is the impact on overall CO<sub>2</sub> emissions?
- What are the main sources of the energy used in households?

## 2. Data Sources

[Statistics | Eurostat \(europa.eu\)](https://ec.europa.eu/eurostat)

This is the website of the statistical office of the European Union – it has a large body of data in multiple different fields which we used to combine to create a story of the energy usage across Europe per household by fuel type and cross-referencing population and GDP.

The choice of which data to use was based on the following:

- Timespan: 2012-2020
- Countries selected: Germany, France, UK, Italy, Spain & Ireland
- Energy sources: Electricity, Solar, Solid fuel, Natural Gas, Oil & Diesel, Biofuel
- Measured in toe (ton of oil equivalent) & MWh per annum
- Greenhouse gases emissions measured in CO<sub>2</sub>-eq

Final energy consumption in households covers the energy used for heating, hot water, cooling, cooking as well as electricity consumption by various electrical appliances. Self-produced electricity is included and counts as consumption of electricity.



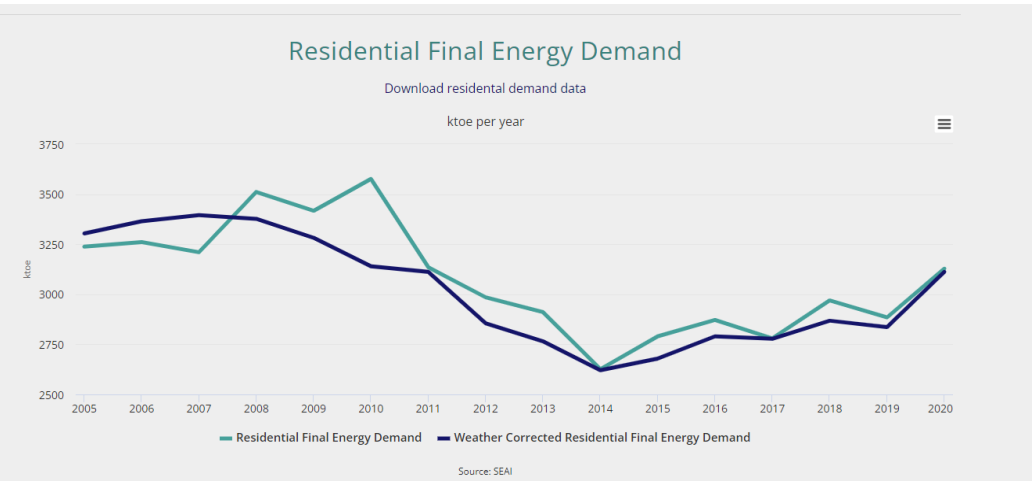
Selecting data source

Data was filtered down by clustering the five countries with the highest consumption and GDP in Europe, based on the data from Eurostat – Ireland was added as a point of comparison. Condensing the information, highlighted in the image below gave us a better understanding of the dataset and helped on the choice of visuals.

| GDP and main components (output, expenditure and income) (online data code: NAMA_10_GDP) |      |              |              |              |              |              | Settings: Default presentation |              |  |
|------------------------------------------------------------------------------------------|------|--------------|--------------|--------------|--------------|--------------|--------------------------------|--------------|--|
| Source of data: Eurostat                                                                 |      |              |              |              |              |              |                                |              |  |
| Table                                                                                    |      | Line         |              | Bar          |              | Map          |                                |              |  |
| IT                                                                                       | TIME | 2012         | 2013         | 2014         | 2015         | 2016         | 2017                           | 2018         |  |
| GEO                                                                                      |      |              |              |              |              |              |                                |              |  |
| European Union - 28 countries (2013-2020)                                                |      | 13 582 975.3 | 13 616 543.5 | 14 895 186.0 | 14 859 516.3 | 14 986 766.4 | 15 436 659.8                   | 15 950 776.2 |  |
| European Union - 15 countries (1995-2004)                                                |      | 12 422 864.3 | 12 519 872.6 | 12 964 282.3 | 13 666 349.7 | 13 762 936.8 | 14 185 126.9                   | 14 522 885.2 |  |
| European Union - 27 countries (from 2007)                                                |      | 11 391 946.4 | 11 528 285.5 | 11 784 825.8 | 12 214 799.8 | 12 552 647.2 | 13 876 869.9                   | 13 529 878.9 |  |
| Euro area - 19 countries (from 2015)                                                     |      | 9 837 864.8  | 9 935 493.8  | 10 168 694.8 | 10 522 291.6 | 10 816 378.4 | 11 223 238.8                   | 11 598 588.6 |  |
| Euro area (EA11-1998, EA12-2001, EA13-2007, EA18-2015)                                   |      | 9 781 556.4  | 9 877 663.9  | 10 132 113.5 | 10 522 291.6 | 10 816 378.4 | 11 223 238.8                   | 11 598 588.6 |  |
| Euro area - 12 countries (2001-2006)                                                     |      | 9 627 228.3  | 9 722 141.2  | 9 948 531.2  | 10 293 128.9 | 10 579 441.4 | 10 970 883.3                   | 11 328 186.1 |  |
| Germany (until 1990 former territory of the FRG)                                         |      | 2 745 318.9  | 2 811 358.8  | 2 927 438.0  | 3 826 188.0  | 3 134 748.0  | 3 267 168.8                    | 3 365 458.8  |  |
| United Kingdom                                                                           |      | 2 111 828.9  | 2 896 338.8  | 2 311 888.2  | 2 644 716.5  | 2 434 119.2  | 2 359 789.9                    | 2 420 897.2  |  |
| France                                                                                   |      | 2 088 884.8  | 2 117 189.9  | 2 149 765.9  | 2 198 432.8  | 2 234 129.0  | 2 297 242.8                    | 2 363 386.8  |  |
| Italy                                                                                    |      | 1 624 358.7  | 1 612 751.3  | 1 627 485.6  | 1 655 355.8  | 1 695 786.8  | 1 736 592.8                    | 1 771 391.2  |  |
| Spain                                                                                    |      | 1 831 184.8  | 1 828 677.8  | 1 832 686.0  | 1 878 892.0  | 1 114 428.0  | 1 162 492.8                    | 1 283 859.8  |  |
| Turkey                                                                                   |      | 683 587.3    | 719 726.6    | 787 888.8    | 777 842.3    | 785 618.9    | 768 497.1                      | 658 544.4    |  |
| Netherlands                                                                              |      | 652 966.0    | 660 463.9    | 671 568.0    | 688 088.8    | 708 337.0    | 738 146.0                      | 773 987.0    |  |
| Switzerland                                                                              |      | 538 439.8    | 536 632.9    | 553 942.2    | 632 778.5    | 628 729.6    | 623 994.0                      | 622 746.8    |  |
| Sweden                                                                                   |      | 438 837.1    | 441 858.7    | 438 833.9    | 455 494.7    | 466 266.5    | 488 825.5                      | 470 673.1    |  |
| Norway                                                                                   |      | 396 523.5    | 393 488.7    | 375 947.3    | 347 632.1    | 333 471.3    | 353 316.4                      | 378 294.3    |  |
| Poland                                                                                   |      | 387 947.8    | 392 318.7    | 488 967.8    | 438 465.8    | 427 891.8    | 467 426.6                      | 497 842.3    |  |
| Belgium                                                                                  |      | 386 174.7    | 392 888.8    | 483 883.3    | 416 781.4    | 438 885.3    | 445 858.1                      | 468 891.7    |  |
| Austria                                                                                  |      | 318 653.8    | 323 918.2    | 333 146.1    | 344 269.2    | 357 688.8    | 369 361.9                      | 385 424.8    |  |
| Denmark                                                                                  |      | 254 578.0    | 258 742.7    | 265 757.8    | 275 017.6    | 283 189.7    | 294 888.2                      | 382 328.7    |  |
| Finland                                                                                  |      | 281 837.8    | 284 321.9    | 286 897.8    | 211 385.8    | 217 518.8    | 226 381.0                      | 233 462.8    |  |
| Greece                                                                                   |      | 188 388.6    | 179 884.4    | 177 236.0    | 176 368.9    | 174 494.2    | 176 983.4                      | 179 557.7    |  |
| Ireland                                                                                  |      | 175 615.4    | 179 457.6    | 195 885.2    | 262 975.9    | 278 285.3    | 297 763.2                      | 326 631.3    |  |

Irish Statistics | SEAI

We also referred to the SEAI website above for data specific to Ireland only.



Separate data contains a more granular view on the fuel types used in homes in Ireland, over a wider timescale including data on CO<sub>2</sub> emissions which will help guide us in our goal of answering our objectives.

Due to differences on the datasets, two power BI reports were created. The reasoning was that the Irish report containing a longer time series and categorised with different fuel types. The comparison would not have been like for like.

## Data cleaning performed

A. Filtered to remove incomplete data such as countries with zero information provided or where only partial data is available. See image below

|                                                         |            |            |            |            |            |            |            |            |            |            |
|---------------------------------------------------------|------------|------------|------------|------------|------------|------------|------------|------------|------------|------------|
| United Kingdom                                          | 37 786,091 | 48 957,945 | 41 458,259 | 35 813,099 | 37 257,018 | 37 970,686 | 36 772,290 | 38 387,568 | 38 134,984 | :          |
| Montenegro                                              | 268,887    | 281,347    | 269,251    | 256,273    | 265,421    | 264,617    | 263,151    | 248,017    | 243,959    | 243,169    |
| North Macedonia                                         | 567,027    | 559,010    | 519,019    | 523,579    | 531,060    | 491,613    | 529,132    | 483,576    | 492,266    | 513,382    |
| Albania                                                 | 496,280    | 509,086    | 580,003    | 556,243    | 532,287    | 496,350    | 492,580    | 510,484    | 504,986    | 541,030    |
| Serbia                                                  | 3 134,547  | 3 122,484  | 2 846,752  | 2 752,774  | 2 826,151  | 2 924,257  | 2 853,554  | 2 838,427  | 2 854,671  | 3 488,669  |
| Turkey                                                  | 20 809,776 | 20 107,463 | 19 807,954 | 19 166,344 | 20 153,339 | 20 711,118 | 22 167,051 | 20 557,145 | 21 563,923 | 23 029,022 |
| Bosnia and Herzegovina                                  | :          | :          | :          | 988,816    | 1 113,839  | 1 139,361  | 1 045,641  | 1 719,424  | 1 686,605  | 1 880,155  |
| Kosovo (under United Nations Security Council Resolu... | 491,178    | 474,234    | 494,163    | 479,943    | 476,388    | 549,413    | 571,465    | 573,484    | 586,881    | 608,818    |
| Moldova                                                 | 1 137,092  | 1 155,360  | 1 147,283  | 1 169,757  | 1 179,946  | 1 233,669  | 1 316,751  | 1 356,862  | 1 237,815  | 1 254,860  |
| Ukraine                                                 | 23 587,848 | 23 454,289 | 23 446,497 | 20 357,427 | 16 551,879 | 17 348,751 | 16 483,165 | 15 986,088 | 14 815,700 | 13 654,325 |
| Georgia                                                 | :          | :          | 1 159,320  | 1 194,979  | 1 201,348  | 1 266,269  | 1 336,297  | 1 227,383  | 1 300,650  | 1 488,174  |

Special value:  
(:) not available

Disclaimer

Image # Source; [Statistics | Eurostat \(europa.eu\)](#) accessed 08/09/22

B. Data was not available for UK in 2020 due to the UK leaving the European Union.

[illegible]

## Why is data cleaning required

Data cleaning is the process of fixing or removing incorrect, corrupted, incorrectly formatted, duplicate, or incomplete data within a dataset. It is important to prepare data for analysis to meet the quality criteria such as validity, uniformity, accuracy, consistency, and completeness. Data cleansing removes unwanted, duplicate, and incorrect data from datasets, and giving accurate results. As in the saying garbage in = garbage out.

Reference; [Data Cleaning: Definition, Benefits, And How-To | Tableau](#) accessed 14/09/2022

## Steps to create Visualisation on PowerBI

### A. Import from data set, select information

### B. Transform the data and rename the headers

|        | 1.2 Country    | 1.2 Column2 | 1.2 Column3 | 1.2 Column4 | 1.2 Column5 | 1.2 Column6 | 1.2 Column7 | 1.2 Column8 | 1.2 Column9 |
|--------|----------------|-------------|-------------|-------------|-------------|-------------|-------------|-------------|-------------|
|        | 1 Countries    | 2012        | 2013        | 2014        | 2015        | 2016        | 2017        | 2018        | 2019        |
| Diesel | Germany        | 59188.124   | 62517.628   | 53648.149   | 54959.824   | 56963.58    | 56777.115   | 55822.36    | 57743       |
|        | France         | 43710.318   | 45888.27    | 37786.96    | 39906.35    | 41834.982   | 41112.358   | 39759.77    | 39619.8     |
| Gas    | United Kingdom | 40857.945   | 41458.259   | 35813.099   | 37257.018   | 37970.686   | 36772.29    | 38307.568   | 38134.9     |
| on     | Italy          | 34348.336   | 34230.62    | 28545.98    | 32494.483   | 32185.101   | 32898.639   | 31905.595   | 31138.3     |
| y      | Spain          | 15530.469   | 14888.925   | 14787.91    | 15269.842   | 14391.695   | 14392.022   | 15178.632   | 14447       |
|        | Ireland        | 2884.811    | 2910.357    | 2623.674    | 2790.719    | 2867.358    | 2775.896    | 2963.959    | 2880.3      |

C. By going to the home menu and selecting use top row as a header

#### D. Remove any incomplete rows of data as per snippet below

| Column1 | Column2        | Column3   | Column4   | Column5   | Column6   | Column7   | Column8   | Column9   |
|---------|----------------|-----------|-----------|-----------|-----------|-----------|-----------|-----------|
| 1       | null           | null      | null      | null      | null      | null      | null      | null      |
| 2       | null           | null      | null      | null      | null      | null      | null      | null      |
| 3       | Countries      | 2012      | 2013      | 2014      | 2015      | 2016      | 2017      | 2018      |
| 4       | Germany        | 2745310   | 2811350   | 2927430   | 3026180   | 3134740   | 3267160   | 3365450   |
| 5       | United Kingdom | 2111028.9 | 2096338   | 2311080.2 | 2644716.5 | 2434119.2 | 2359789.9 | 2420897.2 |
| 6       | France         | 2088804   | 2117189   | 2149765   | 2198432   | 2234129   | 2297242   | 2363306   |
| 7       | Italy          | 1624358.7 | 1612751.3 | 1627405.6 | 1655355   | 1695786.8 | 1736592.8 | 1771391.2 |
| 8       | Spain          | 1031099   | 1020348   | 1032158   | 1077590   | 1113840   | 1161867   | 1203259   |
| 9       | Ireland        | 175615.4  | 179457.6  | 195085.2  | 262975.9  | 270205.3  | 297763.2  | 326631.3  |

Power Query Editor

Transform Add Column View Tools Help

Recent Sources Enter Data Data source settings Manage Parameters Refresh Preview Advanced Editor Manage Query Choose Columns Remove Columns Keep Rows Remove Rows Split Column Group By

Remove Top Rows

Remove Bottom Rows

Remove Alternate Rows

Remove Duplicates

Remove Blank Rows

Remove Errors

|   |                |           |           |
|---|----------------|-----------|-----------|
| 2 | null           | null      | null      |
| 3 | Countries      | 2012      | 2013      |
| 4 | Germany        | 2745310   | 2811350   |
| 5 | United Kingdom | 2111028.9 | 2096338   |
| 6 | France         | 2088804   | 2117189   |
| 7 | Italy          | 1624358.7 | 1612751.3 |
| 8 | Spain          | 1031099   | 1020348   |
| 9 | Ireland        | 175615.4  | 179457.6  |

#### E. Data needed to be appended in order to get total comparisons.

|                                    |  |
|------------------------------------|--|
| Actual Usage Totals                |  |
| All Countries excl total           |  |
| Σ Amount                           |  |
| Country                            |  |
| Energy Type                        |  |
| Energy Type Group                  |  |
| YEAR                               |  |
| Amount GDP                         |  |
| Amount Population                  |  |
| Amount Solar Thermal               |  |
| Amount Solid Fossil Fuel           |  |
| Amounts Conclusion on Germany only |  |
| Amounts Germany                    |  |
| Append All Countries               |  |
| Append1 (3)                        |  |
| Append2                            |  |
| Append2 (2)                        |  |
| Append2 (3)                        |  |

#### F. A grouping of energy types was then generated to create summaries for the types of fuel for filtering by selecting or excluding in the visualisation in some cases.

|                           |  |
|---------------------------|--|
| Actual Usage Totals       |  |
| All Countries Co2 excl... |  |
| Σ Amount                  |  |
| Country                   |  |
| Energy Type               |  |
| Energy Type Gr...         |  |
| YEAR                      |  |

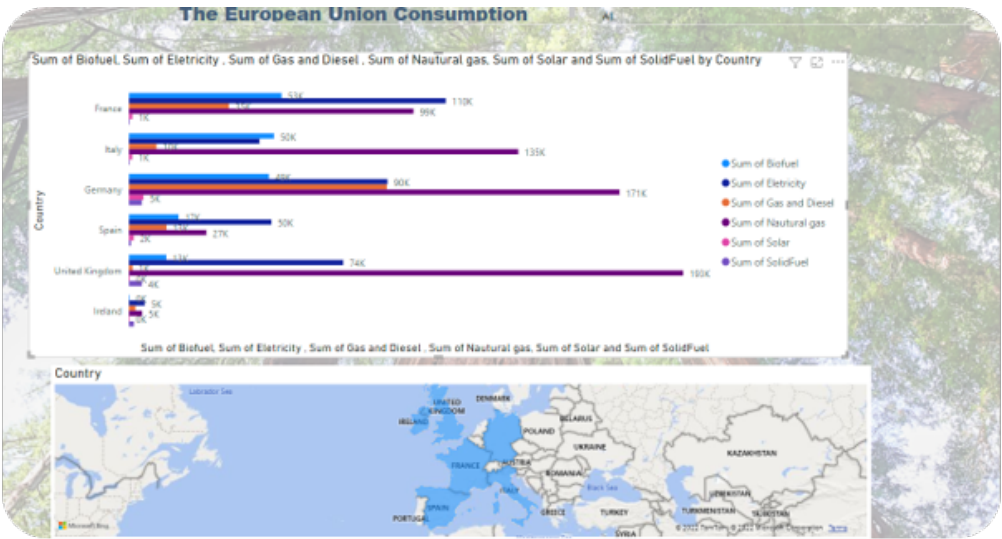
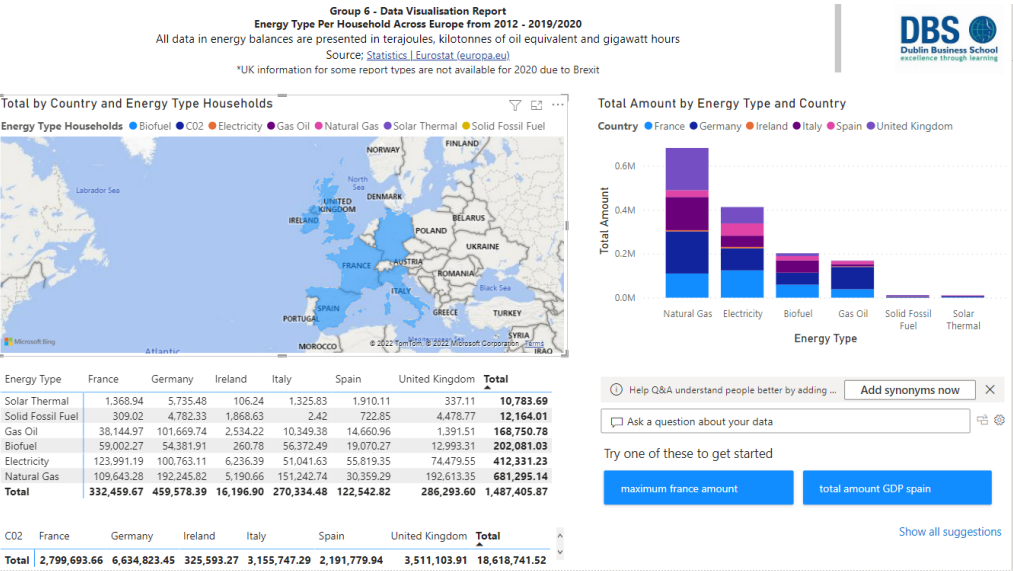
| Energy Type | Country | Amount    | Energy Type Group |
|-------------|---------|-----------|-------------------|
| BioFuel     | France  | 6412.708  | Renewables        |
| Gas Oil     | France  | 4863.439  | Electricity       |
| CO2         | France  | 291912.1  | Ammonia           |
| Gas Oil     | France  | 1596.555  | Electricity       |
| Gas Oil     | France  | 3363.892  | Electricity       |
| Gas Oil     | France  | 3363.201  | Electricity       |
| Natural Gas | France  | 24507.507 | Electricity       |
| Natural Gas | France  | 24503.878 | Electricity       |
| Natural Gas | France  | 20132.208 | Electricity       |
| Natural Gas | France  | 30840.899 | Electricity       |
| CO2         | France  | 339524.88 | Ammonia           |
| Natural Gas | France  | 23373.43  | Electricity       |
| BioFuel     | France  | 6444.86   | Renewables        |
| BioFuel     | France  | 2367.803  | Renewables        |
| Natural Gas | France  | 22630.383 | Electricity       |
| BioFuel     | France  | 4160.474  | Renewables        |

#### G. Once all of the relevant data has been imported into Power BI we then began to create a report of visuals.

Data Visualisations

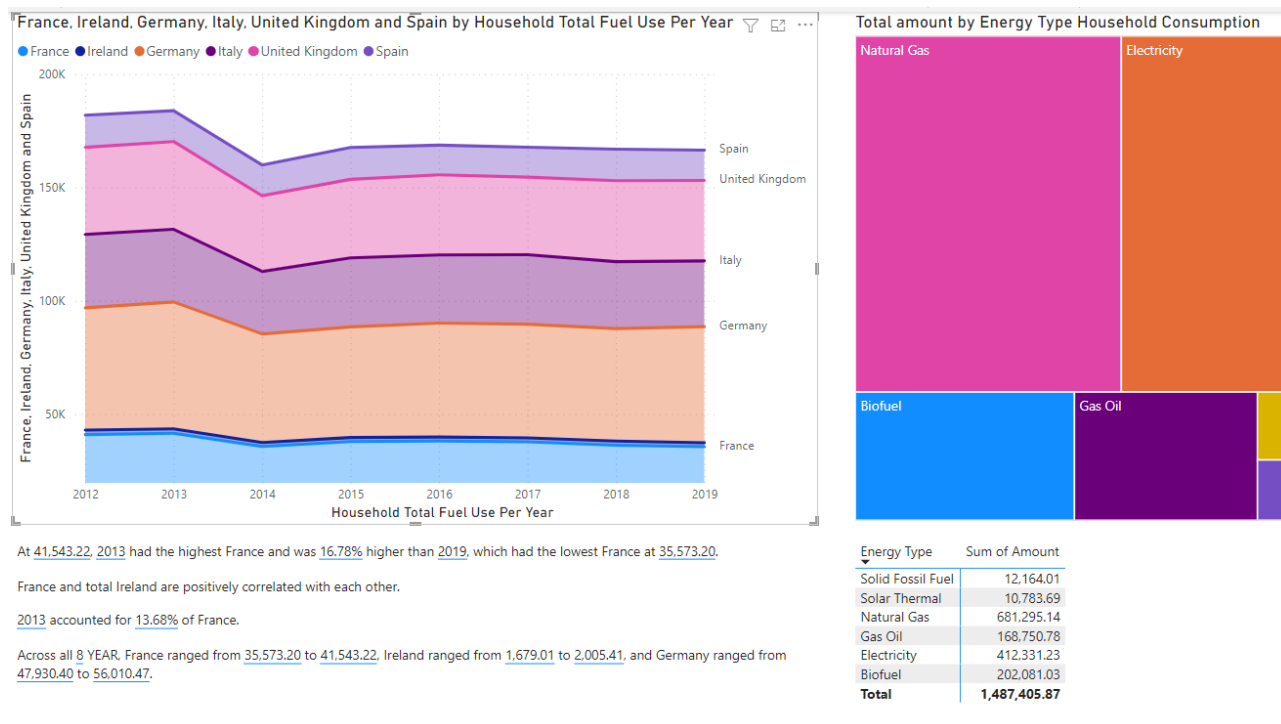
1<sup>st</sup> visualisation – Introduction Summary

An interactive map was created as seen below. Along with a bar chart, a matrix on fuel type, country of the full time series to be analysed. CO<sub>2</sub> was separated by filtering. An interactive question box was inserted for the user to ask questions, answers based on the data.

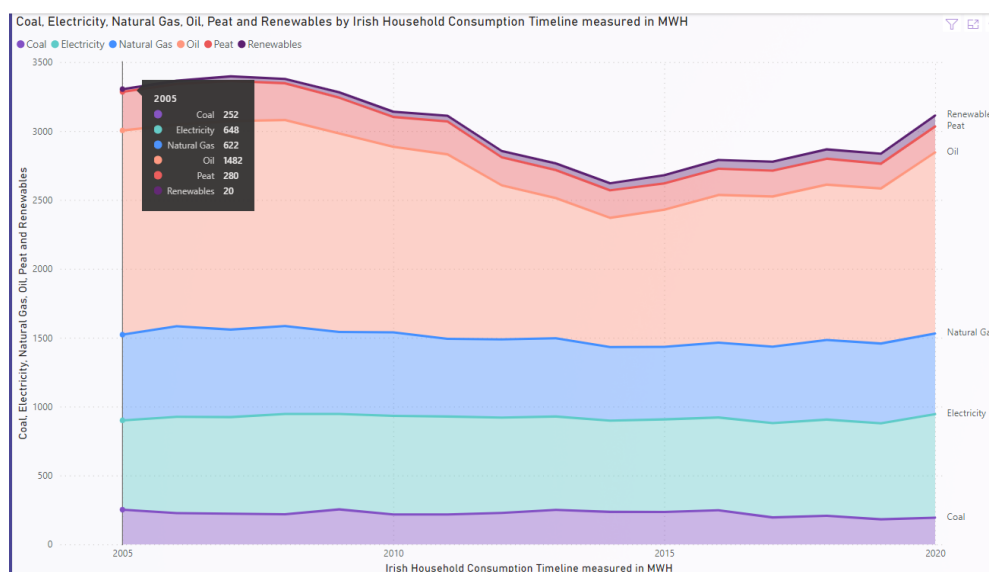


## 2<sup>nd</sup> visualisation – Time Series / Tree map

- A. The next was a stacked area chart. This is best used for a time series.
- B. A tree map was used to breakdown the totals per energy type. This immediately identifies the largest or lowest fuel types along with a matrix of totals.
- C. We could see a decline from 2008 to 2014 across all the countries and this was due to the recession. Growth began to pick up in 2015 / 2016.
- D. A smart narrative function was used with accurate descriptions of the information shown.

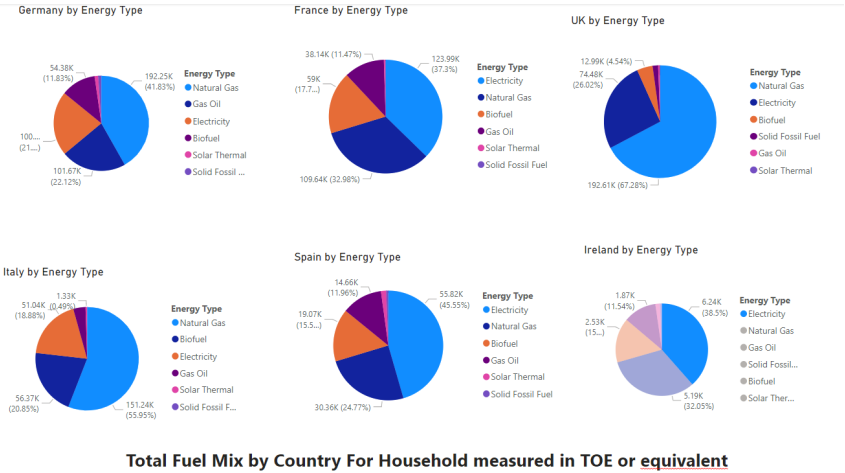


This was repeated for the Irish-only set of data. This mirrors the European changes of decreased from 2008 to 2014 due to the recession.



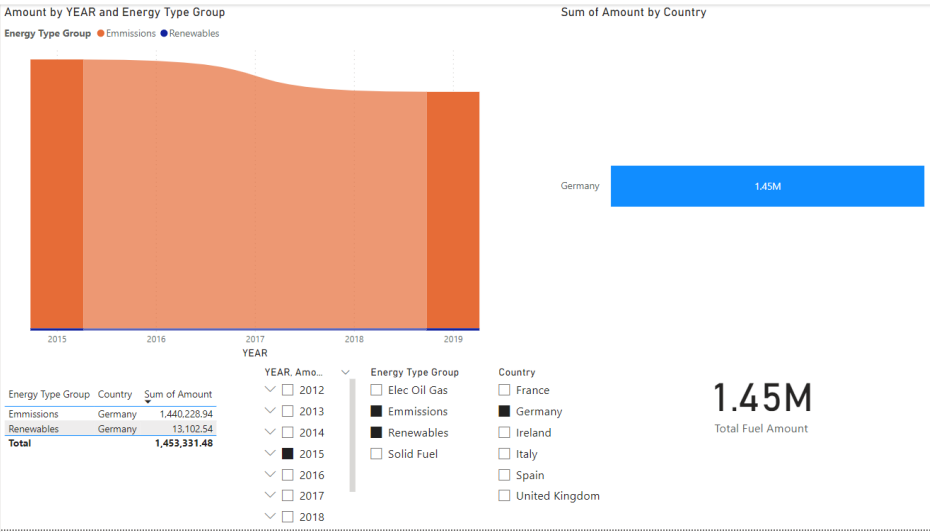
3<sup>rd</sup> visualisation – Country Pies

Pie charts for each country was then used, giving clear visuals of % of fuel types for interpretation.



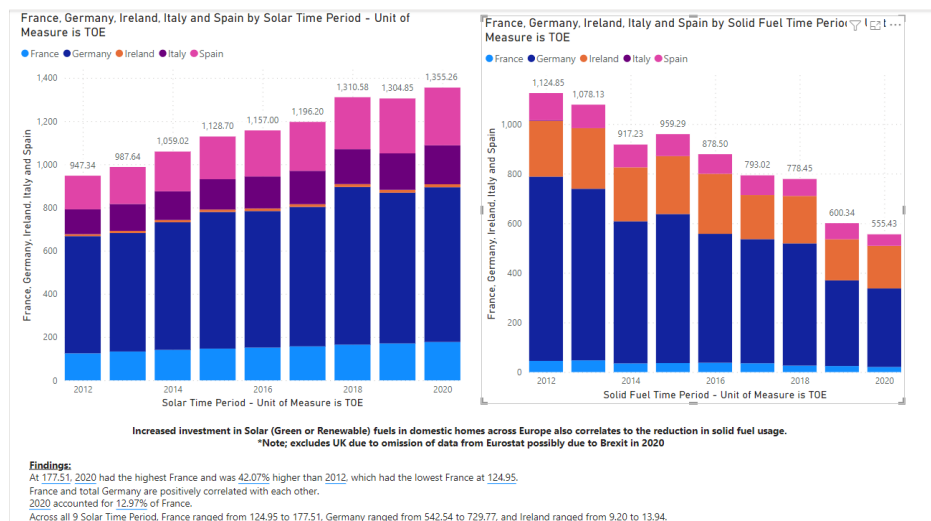
4<sup>th</sup> visualisation – Slices

For interactive purposes, a visual with synced slices of years, energy type, country, along with a bar, a funnel chart and a card were created.



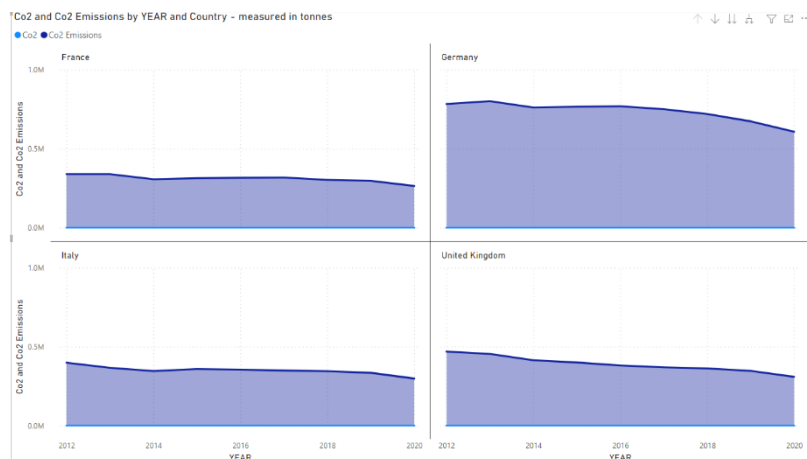
## 5th visualisation – renewable energy v solid fuel

- A line stacked column chart on the summarised for solar thermal was used to analyse the trend for 2012 to 2020. UK was excluded due to lack of data.
- A steady incline could be viewed from year on year with colouring for various countries.
- This was seconded with another on solid fuel usage. Generally, where renewables are seen increasing the higher carbon emission fuels such as coal and peat should in theory reduce as households invest in a greener fuel source. The decline was impressive from initial consumption in 2012 of 1124.85 to 555.43 TOE in 2020 - a decrease over the total time series of 50.6%.

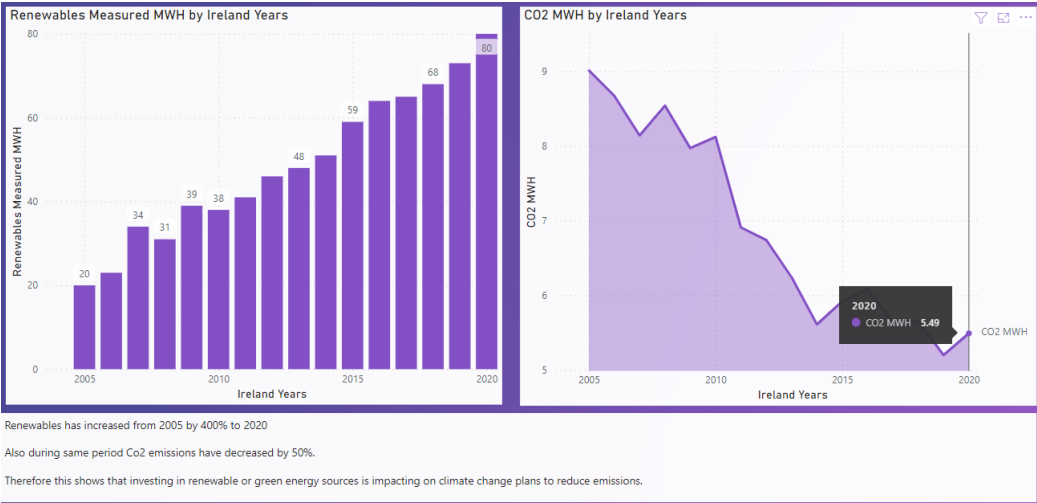


## 6th visualisation – CO2 emissions

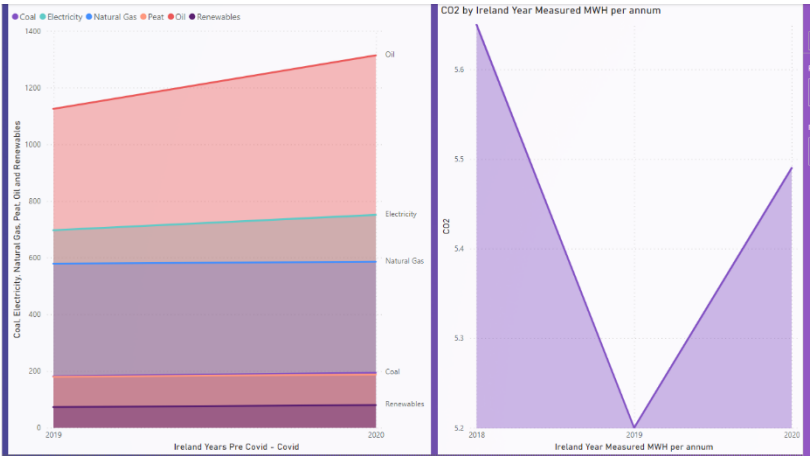
Our objective was to view the impact to CO<sub>2</sub> emissions data, by creating a stacked area chart in quadrant view. All countries seen a dip in 2014, then a slow decline.



For Irish data we compared renewable energy in 2005 to 2020 showing a 400% increase in uptake on consumption of greener energy fuels. Seconded with an area chart of comparable Co2 it can be clearly seen that there was a dip in 2014 but ultimately decreasing by 50%.



As an additional view specific to Ireland we examined any impact on the period prior to Covid to 2020 by an area chart of fuel use and also the CO<sub>2</sub>. There was a slight increase in fuel use from 2019 to 2020. A dip in 2019 CO<sub>2</sub> but then an increase back in 2020. Due to government orders to stay at home, the usage increased during lockdown periods.



## 7th visualisation – Final conclusions

A comprehensive matrix was set up to view the complete data set in sum format broken down into years, countries and energy type with totals using a drill down. Selecting 2012 with 2020 showing the start and end figures.

| YEAR           | Biofuel    | CO2           | Electricity | Gas Oil    | Natural Gas | Solar Thermal | Solid Fossil Fuel | Total         |
|----------------|------------|---------------|-------------|------------|-------------|---------------|-------------------|---------------|
| 2012           | 24,150.92  | 2,294,286.41  | 48,531.10   | 21,161.01  | 84,457.00   | 993.23        | 1,780.52          | 2,475,360.17  |
| France         | 6,807.17   | 339,614.88    | 13,765.11   | 5,332.76   | 14,055.68   | 124.95        | 45.14             | 379,745.67    |
| Germany        | 7,067.88   | 783,913.88    | 11,779.88   | 12,195.83  | 21,618.70   | 542.54        | 742.85            | 837,861.55    |
| Ireland        | 27.81      | 37,000.79     | 698.23      | 445.99     | 599.28      | 9.20          | 224.90            | 39,006.20     |
| Italy          | 6,593.96   | 399,105.28    | 5,972.23    | 1,588.29   | 18,116.55   | 114.93        | 2.42              | 431,493.67    |
| Spain          | 2,484.57   | 265,067.64    | 6,456.41    | 1,453.26   | 3,509.43    | 155.72        | 109.55            | 279,236.57    |
| United Kingdom | 1,169.53   | 469,583.94    | 9,859.24    | 144.89     | 26,557.35   | 45.88         | 655.67            | 508,016.51    |
| 2013           | 25,415.95  | 2,237,399.81  | 48,455.03   | 22,172.12  | 85,548.18   | 1,035.53      | 1,747.76          | 2,421,774.38  |
| 2014           | 21,459.73  | 2,104,590.34  | 45,671.52   | 19,121.33  | 69,872.28   | 1,108.58      | 1,483.09          | 2,263,306.87  |
| 2015           | 22,699.50  | 2,131,191.01  | 46,251.28   | 19,086.08  | 75,303.42   | 1,179.38      | 1,513.06          | 2,297,223.73  |
| 2016           | 21,905.51  | 2,105,116.41  | 46,530.86   | 17,995.62  | 79,333.60   | 1,192.30      | 1,420.76          | 2,273,495.06  |
| 2017           | 22,367.87  | 2,083,827.18  | 46,625.75   | 17,723.67  | 77,730.65   | 1,231.88      | 1,328.16          | 2,250,835.16  |
| 2018           | 22,178.81  | 2,022,459.11  | 46,560.45   | 16,132.01  | 78,654.61   | 1,346.55      | 1,278.68          | 2,188,610.21  |
| 2019           | 22,406.65  | 1,925,694.81  | 46,155.45   | 17,127.25  | 77,900.99   | 1,340.99      | 1,056.56          | 2,091,682.70  |
| 2020           | 19,496.10  | 1,714,176.44  | 37,549.80   | 18,231.69  | 52,494.42   | 1,355.26      | 555.43            | 1,843,859.13  |
| France         | 5,967.90   | 264,729.61    | 13,886.37   | 3,365.20   | 10,753.21   | 177.51        | 20.76             | 298,900.57    |
| Germany        | 5,717.61   | 608,399.41    | 10,922.79   | 12,054.94  | 21,745.67   | 716.30        | 316.48            | 659,873.20    |
| Ireland        | 25.92      | 33,156.29     | 751.33      | 276.92     | 588.56      | 13.94         | 171.78            | 34,984.74     |
| Italy          | 5,955.44   | 298,900.44    | 5,693.18    | 768.59     | 15,932.60   | 180.18        | 0.00              | 327,430.42    |
| Spain          | 1,829.23   | 199,319.27    | 6,296.13    | 1,611.42   | 3,474.39    | 267.33        | 46.41             | 212,844.17    |
| United Kingdom |            | 309,671.42    |             | 154.61     |             |               |                   | 309,826.03    |
| Total          | 202,081.03 | 18,618,741.52 | 412,331.23  | 168,750.78 | 681,295.14  | 10,783.69     | 12,164.01         | 20,106,147.39 |

Due to the vast array of data, we selected the largest consumer of energy – Germany, as a benchmark for our conclusion. A final summary called deviation was calculated based on the changes, turning values into a percentage of deviation using the edit query changing type to a percentage with decimal places.

A table and two smaller matrices were used to show the values and calculations of the findings for CO<sub>2</sub> and Solar thermal. A text box was added alongside to show the overall findings.

## Conclusions

### Conclusions for Europe specific to Germany

- Reduced consumption in the total fuel consumption by 86% in total from 2012-2020
- Largest decrease was in Solid Fuel of 57%
- Increased Solar Thermal (renewables) by 32%
- Overall decrease in CO2 emissions by households of 22%

#### Deviation % of use in Europe from 2012 - 2020. Summary below based on Germany

| Country      | % Change       | Deviation Amount | Energy Type |
|--------------|----------------|------------------|-------------|
| Germany      | -19.10%        | -1,350.27        | Biofuel     |
| Germany      | -1.16%         | -140.89          | Gas Oil     |
| Germany      | -1.16%         | -140.89          | Natural Gas |
| Germany      | -57.40%        | -140.89          | Solid Fuel  |
| Germany      | -7.28%         | 173.76           | Electricity |
| <b>Total</b> | <b>-86.09%</b> | <b>-1,599.16</b> |             |

| Energy Type  | % Change       | Deviation          |
|--------------|----------------|--------------------|
| Co2          | -22.39%        | -175,514.47        |
| <b>Total</b> | <b>-22.39%</b> | <b>-175,514.47</b> |

| Energy Type   | % Change      | Deviation     |
|---------------|---------------|---------------|
| Solar Thermal | 32.03%        | 173.76        |
| <b>Total</b>  | <b>32.03%</b> | <b>173.76</b> |

#### Conclusion - Times Series 2012 to 2020 results of overall changes based on Germany:

- Largest Consumption Users Overall = Germany

- Reduced consumption in fuels by 86% in total

- Largest decrease was in Solid Fuel of 57%

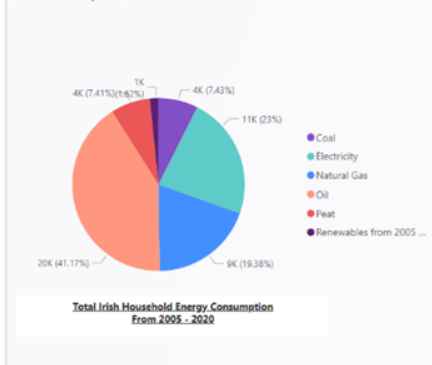
- Increased Solar Thermal (renewables) by 32%

- Overall decrease in CO2 emissions by households of 22%

### Conclusions specific to Ireland

- The data shows a significant reduction in Irish household of high carbon fuels such as coal and peat.
- Peat usage was reduced by 13% over the years.
- An increase in green energy sources of renewables by 74%, example electric energy sources such as heat pumps etc
- This combination in Irish households has a positive impact to the environment by reducing Carbon emissions by 39% of this time.

Coal, Electricity, Natural Gas, Oil, Peat and Renewables from 2005 to 2020



Total Fuel by type from 2005 - 2020

| Coal        | Electricity  | Natural Gas | Oil          | Peat        | Renewables | Year |
|-------------|--------------|-------------|--------------|-------------|------------|------|
| 252         | 648          | 622         | 1482         | 280         | 20         | 2005 |
| 227         | 699          | 657         | 1464         | 295         | 23         | 2006 |
| 223         | 701          | 635         | 1513         | 291         | 34         | 2007 |
| 219         | 728          | 638         | 1495         | 267         | 31         | 2008 |
| 254         | 693          | 595         | 1441         | 259         | 39         | 2009 |
| 217         | 716          | 606         | 1347         | 216         | 38         | 2010 |
| 217         | 711          | 564         | 1339         | 239         | 41         | 2011 |
| 228         | 692          | 568         | 1118         | 203         | 46         | 2012 |
| 251         | 677          | 568         | 1017         | 204         | 48         | 2013 |
| 236         | 662          | 534         | 938          | 200         | 51         | 2014 |
| 235         | 672          | 527         | 995          | 191         | 59         | 2015 |
| 248         | 673          | 543         | 1072         | 190         | 64         | 2016 |
| 196         | 684          | 555         | 1090         | 188         | 65         | 2017 |
| 208         | 698          | 578         | 1127         | 188         | 68         | 2018 |
| 182         | 697          | 579         | 1125         | 180         | 73         | 2019 |
| 194         | 751          | 586         | 1314         | 188         | 80         | 2020 |
| <b>3587</b> | <b>11102</b> | <b>9355</b> | <b>19877</b> | <b>3579</b> | <b>780</b> |      |

Fuel Types



Deviation % of use in Ireland from 2005 to 2020

| Coal   | Electricity | Gas    | Oil     | Peat    | Renew. | Co2     |
|--------|-------------|--------|---------|---------|--------|---------|
| -7.00% | 3.00%       | -5.00% | -12.00% | -13.00% | 74.00% | -39.07% |

#### Findings:

The data shows a significant reduction in Irish household of high carbon fuels such as coal and peat.

Peat usage was reduced by 13% over the years.

An increase in green energy sources of renewables by 74% along with electric energy sources such as heat pumps etc

This combination has a positive impact to the environment by reducing emissions by 39% of this time.

## Final Conclusions

### *Objectives*

Therefore, our objectives set out at the beginning of have been answered.

**Have the five most populous countries in Europe increased the usage of sustainable energy sources over the last decade? What about Ireland?**

Yes, as seen in the 5<sup>th</sup> visual all countries have increased in sustainable energy.

On closer examination Germany being the largest consumer of total energy increased in renewable energy by 32%.

In Ireland, over a longer span of time, seen an increase of 74% .

**What is the impact on overall CO<sub>2</sub> emissions?**

As seen in the 6<sup>th</sup> visual all countries have decreased in CO<sub>2</sub> emissions, with the largest consumer of household energy Germany having a decrease of 22.39% over the time period examined. Ireland reduced the emissions by 39% over the period analysed.

**What can this report be used for and by whom?**

Any householders, consumers, Government agencies etc can view these findings to analyse the shift in green energy and environmental impacts on Co2 emissions based on the fuel used in homes as energy. Governments view data analysis in order to renew or increase any grants needed to encourage uptake on switching to renewable energy.

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Source: [Data Source - Statistics | Eurostat \(europa.eu\)](#)

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